

Table 2. Mock-community benchmark performance at agent and molecule levels

Dataset	Level	Method	Precision	Recall	F1	ARI	Bins	HQ bins
DSMZ dsRNA	Agent	CoCoNet	1.000	0.725	0.840	0.000	67	37
		vRhyme	0.875	0.116	0.205	0.220	4	3
		VAMB	1.000	0.739	0.850	0.000	69	38
		CoCoBin	-	-	-	-	-	-
		CLMB	1.000	0.739	0.850	0.000	69	38
		MetaBAT 2	1.000	0.739	0.850	0.000	69	38
		CONCOCT	0.188	0.986	0.316	0.047	7	0
		MAGE-Bin	1.000	0.754	0.860	0.083	68	39
	Molecule	CoCoNet	1.000	0.913	0.955	0.000	67	59
		vRhyme	0.625	0.116	0.196	0.249	4	1
		VAMB	1.000	0.942	0.970	0.000	69	61
		CoCoBin	-	-	-	-	-	-
		CLMB	1.000	0.942	0.970	0.000	69	61
		MetaBAT 2	1.000	0.942	0.970	0.000	69	61
		CONCOCT	0.145	1.000	0.253	0.009	7	0
		MAGE-Bin	1.000	0.957	0.978	0.400	68	62
DSMZ VANA	Agent	CoCoNet	1.000	0.672	0.804	0.226	57	27
		vRhyme	0.917	0.179	0.300	0.337	5	4
		VAMB	1.000	0.642	0.782	0.000	67	25
		CoCoBin	-	-	-	-	-	-
		CLMB	1.000	0.642	0.782	0.000	67	25
		MetaBAT 2	1.000	0.642	0.782	0.000	67	25
		CONCOCT	0.343	1.000	0.511	0.129	10	1
		MAGE-Bin	1.000	0.701	0.825	0.226	63	29
	Molecule	CoCoNet	0.984	0.821	0.895	0.427	57	43
		vRhyme	0.750	0.179	0.289	0.397	5	3
		VAMB	1.000	0.851	0.919	0.000	67	47
		CoCoBin	-	-	-	-	-	-
		CLMB	1.000	0.851	0.919	0.000	67	47
		MetaBAT 2	1.000	0.851	0.919	0.000	67	47
		CONCOCT	0.224	1.000	0.366	0.043	10	1
		MAGE-Bin	0.985	0.896	0.938	0.427	63	48
Phage mock (Illumina)	Agent	CoCoNet	1.000	0.542	0.703	0.331	17	3
		vRhyme	0.976	0.604	0.746	0.553	7	3
		VAMB	0.979	0.438	0.605	0.099	31	5
		CoCoBin	1.000	0.479	0.648	0.593	3	1
		CLMB	1.000	0.229	0.373	0.011	46	3
		MetaBAT 2	1.000	0.250	0.400	0.019	44	3
		CONCOCT	0.938	0.812	0.871	0.581	13	2
		MAGE-Bin	0.917	0.979	0.947	0.928	6	3
	Molecule	CoCoNet	1.000	0.542	0.703	0.331	17	3
		vRhyme	0.976	0.604	0.746	0.553	7	3
		VAMB	0.979	0.438	0.605	0.099	31	5
		CoCoBin	1.000	0.479	0.648	0.593	3	1
		CLMB	1.000	0.229	0.373	0.011	46	3
		MetaBAT 2	1.000	0.250	0.400	0.019	44	3
		CONCOCT	0.938	0.812	0.871	0.581	13	2
		MAGE-Bin	0.917	0.979	0.947	0.928	6	3
Phage mock (Nanopore)	Agent	CoCoNet	1.000	0.246	0.395	0.102	45	0
		vRhyme	0.955	0.328	0.488	0.182	6	0
		VAMB	0.984	0.344	0.510	0.156	46	0
		CoCoBin	0.893	0.459	0.606	0.303	4	1
		CLMB	1.000	0.246	0.395	0.084	53	0
		MetaBAT 2	0.983	0.344	0.510	0.173	42	0
		CONCOCT	0.820	0.639	0.718	0.386	23	0
		MAGE-Bin	0.869	0.869	0.869	0.748	9	1
	Molecule	CoCoNet	1.000	0.246	0.395	0.102	45	0
		vRhyme	0.955	0.328	0.488	0.182	6	0
		VAMB	0.984	0.344	0.510	0.156	46	0
		CoCoBin	0.893	0.459	0.606	0.303	4	1
		CLMB	1.000	0.246	0.395	0.084	53	0
		MetaBAT 2	0.983	0.344	0.510	0.173	42	0
		CONCOCT	0.820	0.639	0.718	0.386	23	0
		MAGE-Bin	0.869	0.869	0.869	0.748	9	1

Bold indicates the best value within each dataset and level; for Bins, bold is closest to truth. Dashes indicate failed CoCoBin runs.